PLANRS ACADEMY

Detailed Lesson Plan

Heat Transfer — Conduction Demo

Session 1 · 30 Minutes · Beginner

Course Name	Heat Transfer — Conduction Demo
Category	Heat Transfer — Conduction Demo
Subject Area	Environmental Studies (EVS) / Science
Duration	30 Minutes
Session	Session 1 of 1
Depth Level	Beginner
Language	English (Indian English)
Alignment	NEP 2020 · NCERT EVS Scope
Prepared By	Planrs Academy Curriculum Team
Version	1.0 · April 2026

"Riya touched the steel vessel and said — Ouch! Why did it get hot? The fire didn't touch it, did it?"

— The question that starts this lesson

LEARNING OBJECTIVES

LO1	**RECALL**	State that heat can move from one object to another through touch/contact.
LO2	**RECOGNIZE**	Identify everyday Indian examples where conduction takes place — a hot kadhai handle, a metal spoon in hot dal, an iron roti tawa.
LO3	**DISTINGUISH**	Tell the difference between objects that carry heat quickly (metals) and objects that carry heat slowly (wood, plastic, cloth).
LO4	**APPLY**	Explain — in simple words — why a metal spoon gets hot but a wooden ladle does not, when both are placed in the same hot vessel.
LO5	**EXTEND**	Predict which material will get hot faster when given a new scenario (for advanced learners).

LO1–LO3 are MUST-HAVE outcomes. LO4–LO5 are SHOULD-HAVE (extension).

KEY VOCABULARY

Heat	Energy that makes things feel warm or hot
Conduction	Heat moving from one object to another by touching
Conductor	A material that lets heat pass through it quickly
Insulator	A material that slows down or blocks heat
Metal	A material that conducts heat well (iron, steel, copper)
Contact	When two things touch each other

■■ KEY CONCEPTS & INDIAN EXAMPLES

Concept 1: What Is Heat?

Heat is energy that makes things feel warm or hot.

- The sun warming a rooftop terrace in Indian summer
- Hot chai in a steel glass making the glass feel warm
- Feeling warm near a sigri (clay fire pot) in winter

Concept 2: What Is Conduction?

When heat travels from a hot thing to a cooler thing BY TOUCHING, we call it conduction.

- A steel kadhai handle getting hot on the gas stove
- A chapati tawa (iron griddle) becoming hot when placed on flame
- Touching marble floor — it feels cool because marble carries heat away quickly
- Metal bench in school becoming very hot in afternoon sun

Concept 3: Conductors

Materials that allow heat to pass through them easily — they are called conductors.

- Steel — used in dabbas (tiffin boxes), vessels, spoons
- Iron — tawa, wok, railroad tracks
- Copper — traditional vessels used in South Indian kitchens (handi)
- Aluminium — pressure cooker body

Concept 4: Insulators

Materials that slow down or block the movement of heat — they are called insulators.

- Wood — wooden ladle (khunti/karchi), wooden base under hot vessels
- Plastic — handles on kettles and pressure cookers
- Cloth — potholders (kapde ki pakad), mittens used while handling hot vessels
- Rubber — rubber chappal soles (do not transfer heat from hot roads as fast)

Concept 5: Safety in the Kitchen!

Understanding conductors and insulators keeps us safe in the kitchen every day.

- Always hold the wooden/plastic handle of a hot vessel — not the metal body
- Never touch a metal spoon left in hot food with bare hands
- Pot holders and cloth are used in every Indian kitchen for a reason

■ SESSION OUTLINE WITH TIMING

Total Session Duration: 30 Minutes

00:00–03:00 — **HOOK / ENGAGE**
Animated Indian kitchen scene · Riya touches hot vessel · curiosity question

03:00–08:00 — **WHAT IS HEAT? & CONDUCTION**
Short animated explainer · heat = energy · conduction = heat by touch · Indian examples

08:00–15:00 — **CONDUCTORS vs INSULATORS**
Meet Meera the Metal Spoon & Lakdi Bhai the Wooden Ladle · sort kitchen objects

15:00–20:00 — **DRAG & DROP ACTIVITY**
Sort It! — 6 objects dragged into Conductor / Insulator bins · instant feedback

20:00–24:00 — **REAL LIFE + SAFETY**
Why does Grandma use a cloth? · animated scene · safety message · class share

24:00–27:00 — **MINI QUIZ**
3 on-screen MCQs covering LO1, LO2, LO3 · immediate animated feedback

27:00–30:00 — **RECAP + EXIT TICKET**
Summary card · Exit ticket: Name one conductor you have at home

■ SLIDE-BY-SLIDE CONTENT ALIGNMENT

Slide 1	**Title Slide**	Heat Transfer — Conduction Demo · Indian kitchen character (Grandma)
Slide 2	**Hook: Ouch! Why did it get hot?**	Riya story · curiosity question · relatable scenario
Slide 3	**What Is Heat?**	Energy definition · examples: sun, chai, sigri · Indian character
Slide 4	**What Is Conduction?**	Heat moves by touching · 4 Indian examples · discussion question
Slide 5	**Meet Meera & Lakdi Bhai**	Character intro · metal = fast conductor · wood = slow
Slide 6	**Conductors — Heat Carriers**	Steel dabba, Iron tawa, Copper handi, Pressure cooker · 3-card layout
Slide 7	**Insulators — Slowing Heat Down**	Wood, Plastic, Cloth & Rubber · 3-card layout · Grandma character
Slide 8	**Metal Spoon vs Wooden Ladle?**	Explanation slide · safety question · hot vessel illustration
Slide 9	**Safety First in the Kitchen!**	3 safety rules · wooden handle, no bare hands, use cloth
Slide 10	**Let's Play! Sort It!**	Drag-and-drop cue · 6 items listed · Meera & Lakdi Bhai help

Slide 11	**Mini Quiz**	3 questions · Q1: Heat definition · Q2: Conductor · Q3: Safe hold
Slide 12	**Recap — What We Learned**	5 summary bullets with emoji · Indian boy character
Slide 13	**Exit Ticket**	Open question: Name one conductor at home · encourage sharing
Slide 14	**Thank You!**	Closing · Indian family characters (Grandma, boy, girl)

■ MINI QUIZ — QUESTIONS & ANSWERS

Q 1

What is heat?

a) Energy that makes things warm ✓
b) Water
c) Air

■ *Heat is energy. The sun, fire, and hot food all produce heat.*

Q 2

Which material carries heat fast?

a) Metal ✓
b) Wood
c) Cloth

■ *Metal is a conductor — it lets heat pass through quickly.*

Q 3

What should you use to hold a hot vessel?

a) Wooden handle ✓
b) Metal spoon
c) Bare hand

■ *Wood is an insulator — it keeps heat from reaching your hand.*

■ DRAG & DROP SORT ACTIVITY — ANSWER KEY

ITEM	CONDUCTOR / INSULATOR	REASON
Iron nail	**CONDUCTOR ■**	Metal — lets heat pass through quickly
Wooden block	**INSULATOR ■**	Wood — slows down heat significantly
Copper coin	**CONDUCTOR ■**	Copper is an excellent heat conductor
Plastic ruler	**INSULATOR ■**	Plastic — poor conductor of heat
Aluminium foil	**CONDUCTOR ■**	Aluminium metal — conducts heat well
Woollen mitten	**INSULATOR ■**	Wool — traps air, blocks heat transfer

■ TONE & STYLE GUIDE

Voice	Speak TO the child, not AT them. Use 'You', 'Let's', 'We' — inclusive and participatory.
Sentences	Maximum 10–12 words per sentence in narration. Short. Simple. Clear.
Questions	Use rhetorical questions to build curiosity: 'Did you ever touch a hot steel glass?'
Vocabulary	Introduce Conductor, Insulator, Conduction — but always pair with a simple definition and an example immediately.
Positive Only	No 'Wrong!' or 'Incorrect!' sounds. Always say 'Good try! Let's find out together.'
Indian English	Use Indian English conventions. Examples must feel like home to the learner.
Visuals	Bright, warm colours. Flat illustration. Red/orange = hot. Blue = cool. No complex diagrams.
Pacing	3-second pause after each new concept. Interactive break every 5–7 minutes.

■ THINGS TO AVOID

Content

✗ Complex atomic/molecular explanations — this is a beginner session
✗ Introducing radiation or convection — scope is CONDUCTION ONLY
✗ Temperature scales (Celsius/Fahrenheit) — not required at this level
✗ Any scenario glorifying unsafe behaviour around fire or hot objects

Visuals

✗ Generic Western kitchen imagery as primary visuals
✗ Real photographs of children — use illustrations only
✗ White text on yellow background or other low-contrast combinations
✗ More than 3 elements on screen at once

Language

✗ Negative phrasing: 'Wrong!', 'That's incorrect!', 'No.'
✗ Passive voice: Use 'Heat moves' not 'Heat is transferred'
✗ Gendered roles: show all genders in kitchen contexts
✗ Brand names for products — use generic category names

Interaction

✗ Countdown timers for answers — causes anxiety at this age
✗ Requiring typing — use tap/click/drag only
✗ Penalising wrong answers or reducing score
✗ More than 2 steps to complete any single activity

Culture

✗ Representing only one Indian state or region as 'default India'
✗ Food examples that alienate students by dietary restriction
✗ Depicting economic disparity in an insensitive way
✗ Defaulting only to North Indian kitchen references

■ SUCCESS CRITERIA

Learning

✓ ≥ 80% of learners answer all 3 quiz questions correctly (1st or 2nd attempt)
✓ ≥ 75% of learners complete the exit ticket activity
✓ LO1, LO2, LO3 demonstrably covered in assessment data
✓ Average session completion rate ≥ 70%

Engagement

✓ Average time-on-task ≥ 22 minutes (73% of 30-minute session)
✓ Interactive activity replay rate ≤ 3 (high replay may signal confusion)
✓ Drop-off during first 5 minutes < 10% — hook segment must work

Content Quality

✓ All Indian context references pass Regional Diversity Checklist
✓ All audio narration passes Grade 2 Language Level Check
✓ All interactive elements tested on 3+ device types (phone/tablet/desktop)
✓ Subtitles/captions present and accurate for all video segments

Feedback

✓ ≥ 4/5 average teacher rating on content relevance
✓ ≥ 4/5 average teacher rating on age-appropriateness
✓ ≥ 1 spontaneous real-life example reported per student in exit ticket

■ TEACHER DISCUSSION PROMPTS

Use these open-ended questions during or after the session to spark classroom discussion:

Opening	*"Have you ever touched something that felt very hot without being near a fire? What was it?"*
During	*"Why do you think Grandma always uses a cloth or wooden handle to pick up the hot kadhai?"*
Explore	*"If you put a metal spoon and a wooden spoon in hot dal — which one would you be afraid to touch? Why?"*
Safety	*"What would happen if all cooking vessels had metal handles and no wooden or plastic ones?"*
Transfer	*"Look around your home — can you find one conductor and one insulator? Tell us what they are!"*
Exit	*"In one sentence — tell me what conduction means, using your own words."*

■ TAKE-HOME ACTIVITY (NO MATERIALS NEEDED)

Ask a family member to help you find 3 things in your kitchen. For each thing, ask:
'Is this a conductor or an insulator?' Then draw all three and label them. Bring your drawing to class tomorrow and share what you found!

Examples to look for: steel katori, wooden spoon, plastic bottle cap, cloth potholder, copper vessel.

■ PRODUCTION SIGN-OFF CHECKLIST

■	Learning Objectives verified against NCERT EVS scope	*Curriculum Lead*
■	All Indian examples reviewed by regional content reviewer	*Regional Reviewer*
■	Cultural sensitivity checklist signed off	*Content QA*
■	Video narration recorded and reviewed for language level	*Media Team*
■	All interactive activities tested on Android tablet + iOS + Chrome	*QA Team*
■	Quiz questions reviewed against Bloom's Taxonomy Level 1–2	*Curriculum Lead*
■	Safety messaging reviewed by subject matter expert	*SME*
■	Accessibility check: captions, contrast, non-color cues present	*Accessibility Lead*
■	Teacher guide / parent note completed	*Curriculum Lead*

■	Thumbnail and cover art approved by creative lead	*Creative Lead*
■	QA sign-off by Curriculum Lead	*Curriculum Lead*
■	Final upload and metadata tagging on platform complete	*Tech Team*

Planrs Academy · Curriculum Design Team · Version 1.0 · April 2026

'Learning That Belongs to Every Child'